ICT2162 · Management Information Systems

Core Concepts: Data, Process, and Decision Support

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

1. Main idea

Operational systems capture detailed events; analytical systems organize information for trends and comparisons. Decision support combines data with models while keeping assumptions visible. Data quality and access controls affect the reliability of decisions.

2. Example and reasoning

A dashboard showing attendance should define its denominator, date range, and treatment of approved medical absences. A percentage without those rules can mislead even when its arithmetic is correct.

3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

4. Questions for study

- Explain “core concepts: data, process, and decision support” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.